# Evaluation runs: abd (abd-register-civl)

generated 2026-11-05 from brain-data 0799b9755 2026-11-05 11:44:26 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{spec}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 0.4 | 2 | 1 h, 250 GB | 3 | 6 | brain03, 2026-11-01 |
| 3 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-01 |

### stateless: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 0.1 | 2 | 1 h, 250 GB | 3 | 6 | brain03, 2026-11-01 |
| 3 | 3 | 960768 | violation | 6.7 | 4 | 1 h, 250 GB | 20739 | 62214 | brain03, 2026-11-01 |
| 4 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-01 |

**Not run (stateless).** skipped: a smaller size had reached its limit: 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 read-no-writeback bag, 6 correct bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 read-no-writeback bag, 9 correct bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 0.6 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-01 |

### stateful: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 3 | 960768 | violation | 6.3 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-01 |

**Not run (stateful).** skipped: a smaller size had reached its limit: 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 read-no-writeback bag, 6 correct bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 read-no-writeback bag, 9 correct bag.

# offline

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 0.6 | 4 | 10 min, 16 GB | 0.1 | 0.6 | 0.0 | in-memory | brain05, 2026-11-02 |

## offline: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 3 | 960768 | violation | 6.9 | 6 | 10 min, 16 GB | 0.0 | 6.9 | 0.0 | in-memory | brain05, 2026-11-02 |

**Not run (offline).** skipped: a smaller size had reached its limit: 3 bag correct, 4 read-no-writeback bag, 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 read-no-writeback bag, 6 correct bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 read-no-writeback bag, 9 correct bag.

# cross offline, Rust implementation against the Java specification

## cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 3.3 | 160 | 10 min, 16 GB | 3.3 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |

## cross offline, Rust implementation against the Java specification: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 3.3 | 160 | 10 min, 16 GB | 3.3 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 3 | 960768 | violation | 6.9 | 157 | 10 min, 16 GB | 2.9 | 6.9 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Rust implementation against the Java specification).** skipped: a smaller size had reached its limit: 3 bag correct.

# cross online, Java implementation (JMC) against the Rust specification export

## cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 79.5 | 1870 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 3 | | 🕓 (1 h) | | | 1 h, 250 GB | | brain05, 2026-11-02 |

## cross online, Java implementation (JMC) against the Rust specification export: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 3.5 | 559 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 3 | | 🕓 (1 h) | | | 1 h, 250 GB | | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): 2 bag correct, 2 bag read-no-writeback, 3 bag read-no-writeback, 3 bag correct.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 12480 | conforms | 0.7 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 3 | | (1 h) | | | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 544 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 3 | 960768 | violation | 7.9 | 6 | 1 h, 250 GB | 309 | brain01, 2026-11-02 |

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 64.9 | 5608 | 10 min, 16 GB | 64.9 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Rust implementation against its Java twin: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 64.9 | 5608 | 10 min, 16 GB | 64.9 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Rust implementation against its Java twin: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 5.6 | 275 | 10 min, 16 GB | 5.6 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 960768 | 960768 | conforms | 2201.8 | 30793 | 10 min, 16 GB | 2201.8 | 6.9 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 4 bag read-no-writeback, 4 bag correct.
skipped: a smaller size had reached its limit: 3 bag correct.
refused (an input missing at run time): 3 bag read-no-writeback.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 64.9 | 5608 | 10 min, 16 GB | 0.6 | 64.9 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 5.6 | 275 | 10 min, 16 GB | 0.6 | 5.6 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 5.6 | 275 | 10 min, 16 GB | 0.1 | 5.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 960768 | 960768 | conforms | 2201.8 | 30793 | 10 min, 16 GB | 6.9 | 2201.8 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 4 bag read-no-writeback, 4 bag correct.
skipped: a smaller size had reached its limit: 3 bag correct.
refused (an input missing at run time): 3 bag read-no-writeback.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 117.1 | 9140 | 10 min, 16 GB | 117.0 | 1.1 | 0.1 | in-memory | brain02, 2026-11-02 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 117.0 | 9140 | 10 min, 16 GB | 117.0 | 0.2 | 0.0 | in-memory | brain02, 2026-11-02 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 10.3 | 1697 | 10 min, 16 GB | 10.3 | 0.2 | 0.0 | in-memory | brain02, 2026-11-02 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: 3 bag correct.
refused (an input missing at run time): 3 bag read-no-writeback.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 117.0 | 9140 | 10 min, 16 GB | 1.1 | 117.0 | 0.0 | in-memory | brain02, 2026-11-02 |

### context-free twins offline, Java port against the Rust implementation: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 10.3 | 1697 | 10 min, 16 GB | 1.1 | 10.3 | 0.0 | in-memory | brain02, 2026-11-02 |

### context-free twins offline, Java port against the Rust implementation: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 10.3 | 1697 | 10 min, 16 GB | 0.2 | 10.3 | 0.0 | in-memory | brain02, 2026-11-02 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 3 bag correct, 3 bag read-no-writeback.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 116.1 | 9686 | 1 h, 250 GB | | brain02, 2026-11-02 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 11.1 | 2359 | 1 h, 250 GB | | brain02, 2026-11-02 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **read-no-writeback**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 10.9 | 2218 | 1 h, 250 GB | | brain02, 2026-11-02 |
| 3 | 960768 | 960768 | conforms | 3176.8 | 36433 | 1 h, 250 GB | | brain02, 2026-11-02 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** skipped: a smaller size had reached its limit: 3 bag correct, 3 bag read-no-writeback.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 12480 | conforms | 1.1 | 8 | 1 h, 250 GB | | brain02, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair read-no-writeback, model bag, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12480 | 544 | conforms* | 0.1 | 4 | 1 h, 250 GB | | brain02, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair read-no-writeback, model bag, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 544 | 544 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain02, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** skipped: a smaller size had reached its limit: 3 bag correct, 3 bag read-no-writeback.